RESEARCH PROJECT • ASSIGNMENT 1

Foundations: Corpus, Noise, and WordPiece

Build the reproducible clean/noisy foundation that the semester research project will use.

TEAM	PHASE	EVIDENCE	SUBMIT
2 students	1 — Foundations	20 paired examples	Report + code package

CORE RULE Apply character noise to raw UTF-8 text before tokenization; anything after WordPiece is token-level noise. Use the same BERT checkpoint, paired tokenizer, examples, and settings in both branches.

1. Purpose

Assignment 1 establishes the data, corruption, tokenization, and verification conventions for the semester study of robust NLP under character noise. Your goal is an inspectable foundation—not a final model result. Each output should make later clean/noisy comparisons reproducible.

2. Assignment question

QUESTION Can the team generate deterministic character-noise variants, measure their edit distance correctly, and show how the fixed BERT WordPiece tokenizer changes on matched clean/noisy text?

Learning outcomes

- Audit a text corpus and document its source, split, encoding, and sampling procedure.
- Implement deterministic insertion, deletion, and substitution at the character level.
- Use unit-cost Levenshtein distance and verify it with known-answer tests.
- Inspect the WordPiece tokenizer paired with a BERT checkpoint; do not train a replacement vocabulary.
- Produce matched clean/noisy evidence with a reproducible configuration and ethical-use record.

SEMESTER CONNECTION Later assignments will add representation, performance, and scale analyses. Power-law behavior is a possible research hypothesis—not an axiom and not a required Assignment 1 result.

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Data and team responsibilities

Use a bounded corpus slice, fixed configuration, and a shared integration contract.

3. Data protocol

Decision	Assignment 1 requirement
Corpus	WikiText-2; record dataset version and source
Scope	Choose a manageable, deterministic slice and record selection rule, count, and checksum
Splits	Develop on training data, use validation for checks, and do not inspect test outcomes to revise code
Encoding	Use UTF-8; preserve raw files; transform in code; save derived files separately; document whitespace and Unicode handling

4. Individual responsibilities

Role	Required work and evidence
Student A — corpus + noise	Audit character categories; implement configurable insertion, deletion, and substitution; fix the seed; save operation, location, source character, replacement, and clean/noisy IDs.
Student B — tokenizer + baseline	Load the checkpoint's paired WordPiece tokenizer; record library/version, checkpoint, casing/normalization, special tokens, vocabulary source/checksum, and maximum length. Add one sentence distinguishing frequency-based BPE merges from WordPiece's likelihood/association-style rule and ## convention.
Both — integration	Agree on IDs and file schema; review unit tests; create and inspect the same 20 paired examples; sign the shared acceptance checklist.

5. Integration contract

Noisy branch — compare with an unchanged clean branch



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Minimum Edit Distance and verification

Use non-negative unit costs and prove correctness with small, known-answer tests.

6. Unit-cost Levenshtein standard

Operation	Cost	Interpretation
Insertion	1	Add one character
Deletion	1	Remove one character
Substitution	1	Replace one character with another

RECURRENCE $D[i,j] = \min(D[i-1,j] + 1, D[i,j-1] + 1, D[i-1,j-1] + [s_i \neq t_j])$; initialize $D[i,0] = i$ and $D[0,j] = j$. Every cell is a non-negative prefix distance.

Required known-answer tests

Source	Target	Expected MED	Purpose
same	same	0	identity
ε (empty)	nlp	3	empty-string boundary
kitten	sitting	3	mixed substitutions/insertion
flaw	lawn	2	deletion/insertion
intention	execution	5	lecture reference case

7. Paired evidence package

Field	Minimum record for each of 20 pairs
Identity + text	example ID, source split/index or hash, clean text, noisy text
Corruption	seed, operation, intensity, edit location, operation log
Distance	MED and normalized MED = MED / max(1, clean character count)
Tokenization + comparison	clean/noisy pieces, counts, continuation pieces, [UNK], truncation, token-count change, short observation

ACCEPTANCE The same seed reproduces the same noise and log; intensity 0 returns the source and MED 0; recorded edits match output; clean/noisy cases share tokenizer settings; and at least one case shows a meaningful WordPiece change without mislabeling BPE.

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Deliverables and report standard

Submit a compact package whose claims can be traced back to code and saved evidence.

8. Submission package

Artifact	Required content
Technical report (PDF)	3–5 pages plus references; concise foundation memo, not the final ACL manuscript
Repository or Colab	runnable source/notebook, tests, configuration, environment notes, and README
Corpus audit	data card with source/version, license or terms, encoding, split use, slice rule, counts, checksum
Evidence + tests	20 paired records in readable and CSV/JSON form; saved MED and deterministic test output
Contribution record	brief statement of individual work and shared integration decisions

9. Report structure

Section	Minimum content
1. Objective	assignment question and role in the semester project
2. Data	corpus source, slice, splits, UTF-8 handling, category statistics
3. Methods	noise design, fixed tokenizer/checkpoint, MED definition, configuration
4. Verification + findings	tests, 20-pair summary, diagnostics, representative examples; no causal overclaim
5. Limitations + ethics	coverage limits, possible misuse, authorized-testing boundary, next steps

10. Reproducibility checkpoint

SUBMISSION STANDARD Record environment/library versions, data and configuration checksums, tokenizer/checkpoint settings, run order, seed, and artifact paths. A classmate should be able to recreate the 20-pair table by following the README.

STARTING POINTS Devlin et al. (2019), BERT; Jurafsky and Martin, Speech and Language Processing; Kaplan et al. (2020), Scaling Laws (later-hypothesis background only).

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Ethics, checklist, and assessment

Treat character perturbation as a dual-use method and communicate results responsibly.

11. Dual-use, ethics, and responsible testing

SAFETY BOUNDARY Character noise can be used defensively to study robustness, but it can also obfuscate harmful instructions or evade filters. This BERT study is not a prompt-injection exercise. Do not probe third-party systems or attempt filter evasion.

- Use only instructor-approved data, models, and systems that you are authorized to test.
- Use benign examples; do not create or publish operational attack strings, bypass recipes, or sensitive targets.
- Frame security-relevant observations around detection, normalization, robustness, and mitigation.
- Document data licensing, privacy, bias, compute use, limitations, and any unexpected behavior.

12. Submission checklist

☐ Raw files remain unchanged; derived files and UTF-8 handling are documented.
☐ Character noise occurs before WordPiece tokenization.
☐ The BERT checkpoint and its paired tokenizer are fixed across clean/noisy branches.
☐ Insertion, deletion, and substitution each cost 1 in MED calculations.
☐ All known-answer and deterministic tests pass and their outputs are saved.
☐ The package contains 20 traceable clean/noisy pairs and tokenizer diagnostics.
☐ README, environment, seed, configuration, and contribution record are complete.
☐ The ethics section states authorized-use limits and dual-use implications.

13. Assessment criteria

Criterion	Evidence of strong work	Weight
Data integrity + audit	traceable, bounded, encoded and documented corpus slice	20%
Noise implementation	deterministic functions, manifest, and valid controls	20%
Tokenizer baseline	correct fixed WordPiece configuration and useful diagnostics	15%
MED + verification	unit-cost standard, boundaries, known-answer tests	15%
Integration + evidence	paired schema, 20 inspectable records, consistent settings	15%
Report + responsibility	clear claims, reproducibility, limitations, ethics	15%